Vector Calculus and Complex Analysis Homework 9

1. If u and v are expressed in terms of polar coordinates (r, θ), show that the Cauchy-Riemann equations can be written in the form

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

Solution

1. $z = re^{i\theta} \Longrightarrow x = r\cos\theta$ **and** $y = r\sin\theta$ **and**

$$f(z) = u(x(r,\theta), y(r,\theta)) + iv(x(r,\theta), y(r,\theta))$$

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta$$

Similar applications of the chain rule yield

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}(-r\sin\theta) + \frac{\partial u}{\partial y}r\cos\theta$$
$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta$$
$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x}(-r\sin\theta) + \frac{\partial v}{\partial y}r\cos\theta$$

Replace the partial derivatives on the right sides of the equations for $\frac{\partial u}{\partial r}$ and $\frac{\partial v}{\partial r}$ by their Cauchy-Riemann counterparts to obtain:

$$\frac{\partial u}{\partial r} = \frac{\partial v}{\partial y}\cos\theta - \frac{\partial v}{\partial x}\sin\theta = \frac{1}{r}\frac{\partial v}{\partial \theta}$$
$$\frac{\partial v}{\partial r} = -\frac{\partial u}{\partial y}\cos\theta + \frac{\partial u}{\partial x}\sin\theta = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

2(a) Evaluate $(-1+i)^i$, i^i and $(-1-i)^i$.

(b) As we all know, $(-1+i)\,i=-1-i$. Is $\left(\text{principal value of } (-1+i)^i\right)\times\left(\text{principal value of } i^i\right)$ $=\left(\text{principal value of } (-1-i)^i\right)$ correct?

(c) Is there any equality in $(-1+i)^{2i}$, $\left((-1+i)^2\right)^i$, $(-1+i)^i\times(-1+i)^i$?

Solution

(a) $(-1+i)^i=e^{i\log(-1+i)}=e^{i\left(\mathrm{Log}\sqrt{2}+i\pi(2k+3/4)\right)}=e^{(-3/4-2k)\pi+i\mathrm{Log}\sqrt{2}}$

$i^i=e^{i\log i}=e^{i(i\pi(2k+1/2))}=e^{(-1/2-2k)\pi}$

$(-1-i)^i=e^{i\log(-1-i)}=e^{i\left(\mathrm{Log}\sqrt{2}+i\pi(2k-3/4)\right)}=e^{(3/4+2k)\pi+i\mathrm{Log}\sqrt{2}}$

(b) No. It is because $e^{(-3/4)\pi+i\mathrm{Log}\sqrt{2}}e^{(-1/2)\pi}=e^{(-5/4)\pi+i\mathrm{Log}\sqrt{2}}\neq e^{(3/4)\pi+i\mathrm{Log}\sqrt{2}}$

(c) None of them are equal.

$(-1+i)^{2i}=e^{2i\log(-1+i)}=e^{2i\left(\mathrm{Log}\sqrt{2}+i\pi(2k+3/4)\right)}=e^{(-3/2-4k)\pi+i\mathrm{Log}2}$

$\left((-1+i)^2\right)^i=\left(1-2i+i^2\right)^i=\left(-2i\right)^i=e^{i\log(-2i)}=e^{i(\mathrm{Log}2+i\pi(2k-1/2))}=e^{(1/2-2k)\pi+i\mathrm{Log}2}$

$(-1+i)^i\times(-1+i)^i=e^{(-3/4-2k)\pi+i\mathrm{Log}\sqrt{2}}\times e^{(-3/4-2l)\pi+i\mathrm{Log}\sqrt{2}}=e^{(-3/2-2m)\pi+i\mathrm{Log}2}$

3. Compute the following complex integrals:

(a) $\int_\Gamma \left(3z^2-5z+i\right)dz$ along the line segment from $z=i$ to $z=1$.

(b) $\int_\Gamma e^z\,dz$ along the upper half of the circle $|z|=1$ from $z=1$ to $z=-1$.

(c) $\int_\Gamma 1/z\,dz$ for any contour in the right half-plane from $z=-3i$ to $z=3i$.

(d) $\int_\Gamma \csc^2 z\,dz$ for any closed contour that avoids the points $0, \pm\pi, \pm2\pi, \ldots$.

(e) $\int_C (x-2xyi)dz$ over the curve $C: z=t+it^2, 0\le t\le 1,$ where $x=\mathrm{Re}\,z$, $y=\mathrm{Im}\,z$.

(f) $\int_C (|z-1+i|^2-z)dz$ along the semicircle $z=1-i+e^{it}, 0\le t\le\pi.$

Solution

(a) $\int_\Gamma \left(3z^2 - 5z + i\right)\, dz$ **along the line segment from** $z = i$ **to** $z = 1$

Steps:

1. The integrand $f(z) = 3z^2 - 5z + i$ is a polynomial and analytic everywhere.
2. Compute the antiderivative:

$$\int f(z)\, dz = z^3 - \frac{5}{2}z^2 + iz + C.$$

3. Evaluate at bounds:

$$\left[z^3 - \frac{5}{2}z^2 + iz\right]_i^1 = \left(-\frac{3}{2} + i\right) - \left(\frac{3}{2} - i\right) = -3 + 2i.$$

Answer:

$$\boxed{-3 + 2i}$$

(b) $\int_\Gamma e^z\, dz$ **along the upper half of** $|z| = 1$ **from** $z = 1$ **to** $z = -1$

Steps:

1. $f(z) = e^z$ is entire (analytic everywhere).
2. Antiderivative: $\int e^z\, dz = e^z + C$.
3. Evaluate:

$$[e^z]_1^{-1} = e^{-1} - e^1.$$

Answer:

$$\boxed{\frac{1}{e} - e}$$

(c) $\int_\Gamma \frac{1}{z}\,dz$ for any contour in the right half-plane from $z=-3i$ to $z=3i$

Steps:

1. $f(z)=\frac{1}{z}$ is analytic in $\mathrm{Re}(z)>0$.
2. Antiderivative: $\mathrm{Ln}\,z$ (principal branch).
3. Evaluate:

$$\mathrm{Ln}(3i)-\mathrm{Ln}(-3i)=\left(\ln 3+\frac{\pi}{2}i\right)-\left(\ln 3-\frac{\pi}{2}i\right)=\pi i.$$

Answer:

$$\boxed{\pi i}$$

(d) $\int_\Gamma \csc^2 z\,dz$ for any closed contour avoiding $z=n\pi$ ($n\in\mathbb{Z}$)

Steps:

1. $\csc^2 z$ has singularities at $z=n\pi$.
2. Antiderivative: $-\cot z$.
3. By Cauchy's theorem, the integral over a closed contour avoiding singularities is zero.

(f)

1. **Parametrization:**
   $z(t)=1-i+e^{it}$, $dz=ie^{it}dt$.
2. **Integrand simplification:**

$$|z-1+i|^2=1,\quad \text{thus}\quad f(z)=1-z=i-e^{it}.$$

3. **Integral setup:**

$$\int_0^\pi (i-e^{it})\cdot ie^{it}\,dt=\int_0^\pi (i^2e^{it}-ie^{2it})\,dt=\int_0^\pi (-e^{it}-ie^{2it})\,dt.$$

4. **Term-wise integration:**
   - $\int -e^{it}\,dt=-ie^{it}\Big|_0^\pi=-i(-1-1)=-2i.$
   - $\int -ie^{2it}\,dt=-\frac{1}{2}e^{2it}\Big|_0^\pi=0.$

Correct result:

$$\boxed{-2i}$$

4. Evaluate each of the following.

(a) $\log i$

(b) $\log(1-i)$

(c) $\text{Log}(-i)$

(d) $\text{Log}(\sqrt{3}+i)$

Solution

a. $i\left(\frac{\pi}{2}+2k\pi\right),\ k=0,\pm1,\ldots$

b. $\frac{1}{2}\text{Log}\,2+i\left(-\frac{\pi}{4}+2k\pi\right),\ k=0,\pm1,\ldots$

c. $-\frac{\pi}{2}i$

d. $\text{Log}\,2+i\frac{\pi}{6}$

5. Suppose the contour joining $-i$ and i lies in the complex plane excluding the origin and the negative real axis. Evaluate

$$\int_{-i}^{i} z^{1/2}dz$$

where the principal branch of $z^{1/2}$ $(-\pi < Arg\ z \le \pi)$ is taken.

$$\begin{aligned}
\int_{-i}^{i} z^{1/2}dz &= \frac{2}{3}\left[z^{3/2}\right]_{-i}^{i} \\
&= \frac{2}{3}\left[i^{3/2}-(-i)^{3/2}\right] \\
&= \frac{2}{3}\left[e^{(3/2)\text{Ln}i}-e^{(3/2)\text{Ln}(-i)}\right] \\
&= \frac{2}{3}\left[e^{(3i\pi/4)}-e^{(-3i\pi/4)}\right] \\
&= \frac{2}{3}\left[\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)-\left(\cos\frac{-3\pi}{4}+i\sin\frac{-3\pi}{4}\right)\right] \\
&= \frac{2\sqrt{2}i}{3}
\end{aligned}$$